

PRATEEK GUPTA

Agricultural and Food Engineering (B.Tech + M.Tech Dual 5Y) | 22AG36022

✉ Email | ☎ +91-6378861223 | 🌐 Website | 💼 LinkedIn | 🐙 GitHub

Education

Degree	Institute	Board/University	CGPA/Percent	Year
B.Tech + M.Tech, AG	IIT Kharagpur	IIT Kharagpur	Ongoing	2022–2027
Class XII	Delhi Public School, Bharatpur	CBSE	91.4%	2022
Class X	T.M. International, Bharatpur	CBSE	96.6%	2020

Research Publication

Topological Analysis of Plant Structure Using Persistent Homology for Genotype Classification Nov 2025
ACM Conference on Computer Vision, Graphics and Image Processing

- Developed a plant phenotyping framework using **Persistent Homology** and skeleton-based features for structural analysis
- Combined **topological and color features** to improve genotype classification performance using **RF** and **SVM** methods
- Enhanced the feature space with **900-D** persistence images and **20+** PH statistics for improved genotype classification
- Achieved **94%** accuracy using a unified **skeleton + PH setup**, outperforming baseline models by **6%** overall in evaluation

COSINE: Federated Semantic Intrusion Detection with Calibrated Gated Rejection Jan 2026
Manuscript under review

- Designed a **semantic, open-set federated IDS** replacing closed-set classification with gated rejection across **5+ classes**
- Embedded network flows and **LLM-derived attack descriptions** into a shared **cosine space** for class-wise reasoning use
- Calibrated **per-class similarity thresholds** and margin gaps to handle **non-IID client heterogeneity** in practice reliably
- Achieved **97.8%** robust accuracy with **92%+** coverage on the **NSL-KDD** intrusion detection benchmark dataset used

Internships and Projects

Data Science Intern | *Algostrauss* May 2024 – July 2024

- Achieved an **F1-score of 0.87** in predicting customer churn using RF, XGBoost & LogReg through targeted refinement
- Optimized hyperparameters using **Optuna**, achieving a **5%** increase in predictive accuracy by enhancing model stability
- Engineered **100+** features from customer, transaction, event, and engagement data, achieving a **10%** lift in performance
- Built interactive **Tableau** dashboards to visualize key business metrics for stakeholders, enabling clearer data-driven views

Personalized Credit Card Offer Ranking System | *AMEX Hackathon* June 2025 – July 2025

- Built an offer-ranking system using LightGBM and LambdaRank, raising **MAP@7** from **0.32 to 0.44** through refinements
- Engineered **250+** features from **100M+** customer, transaction, and event rows to support improved ranking optimization
- Applied **SelectKBest** on **600+** raw and engineered features to retain the top **250 predictors**, improving MAP@7 stability
- Tuned **20+** hyperparameters with Optuna for a **15%** lift in recommendation accuracy across multiple customer segments

Neural PID-ACC Controller | *Academic Project* November 2025

- Replaced the **PID controller** in an ACC system with a **compact neural network** model trained on PID data for control
- Built and trained a **TinyNN 6→8→1** model with **INT8** quantization **<40 KB** giving **<5 ms** inference on Raspberry Pi
- Logged and processed **680** sensor samples (distance, velocity, PID error) to train a supervised **PWM-prediction model**
- Deployed the ML controller on Raspberry Pi and integrated it into the motor loop, yielding stable **PWM outputs** reliably

Technical Skills

Languages	Python SQL Arduino IDE MATLAB Bash/Shell Scripting CSS HTML C C# C++ Rust
ML/DL Libraries	PyTorch TensorFlow Scikit-learn Pandas NumPy Matplotlib Seaborn Prophet Statsmodels
Core Competencies	Machine Learning Deep Learning PCA GNN PINN Time Series Analysis Statistical Analysis RL
NLP Architectures	Encoder-Decoder Attention Transformers Transfer-Learning BERT GPT LLMs Llama Mistral
Tools & Software	Git GitHub Jupyter Anaconda Tableau FreeCAD Microsoft Office Suite CUDA Label Studio
Frameworks	MLflow Docker LangChain Hugging Face OpenCV Autoencoders spaCy PyTorch Lightning
Statistics	Hypothesis Testing Probability Theory A/B Testing Statistical Inference Stationarity Testing

Relevant Coursework & Certifications

- Key Courses:** Programming & Data Structures| DSA, Embedded ML| Probability & Statistics| Intelligent Control
- Certification:** Advanced Machine Learning| Mathematics for Machine Learning Specialization| Bayesian Statistics
- Academics:** Linear Algebra| Partial Differential Equation| Edge Computing| Spatial Informatics| AI in Economics

Awards and Achievements

- Gold Medal – IIT Kharagpur **General Championship Data Science Competition**, securing 1st place among the institute
- Gold Medal in **General Championship (Nukkad)** at IIT Kharagpur, Jan 2025 winning the institute's top cultural event
- Represented district at the **State Level Lawn Tennis Tournament**, securing 1st Place in District with strong match